Putnam and Beyond — Solutions

Contents

Chapter 1

Methods of Proof

1.1 Argument by Contradiction

Problem 1. Prove that $\sqrt{2}+\sqrt{3}+\sqrt{5}$ is an irrational number.

Solution. Assume the contrary, that $\sqrt{2}+\sqrt{3}+\sqrt{5}=r$, where r is rational. Then $\sqrt{2}+\sqrt{3}=r-\sqrt{5}$. Squaring both sides and simplifying, we get $2\sqrt{6}=r^2-2r\sqrt{5}$. Then it follows that $2\sqrt{6}+2r\sqrt{5}=q$ where $q=r^2$ is rational. Squaring both sides again, we get $24+20r^2+8r\sqrt{30}=q^2$, hence $\sqrt{30}$ is rational. Write $\sqrt{30}=a/b$ in lowest terms where a,b are positive integers. Then $30b^2=a^2$, so a is divisible by 2, so we can write $30b^2=4(a/2)^2$. Then $15b^2=2(a/2)^2$, so b is also divisible by 2, so the fraction was not in lowest terms, a contradiction. We conclude that the initial assumption was false, and therefore $\sqrt{2}+\sqrt{3}+\sqrt{5}$ is irrational.

Problem 2. Show that no set of nine consecutive integers can be partitioned into two sets with the product of the elements of the first set equal to the product of the elements of the second set.

Solution. Asssume the contrary, that there exists a set of nine consecutive integers $\{n,n+1,\dots,n+8\}$ that can be partitioned into two sets A and B such that the product of the elements of A equals the product of the elements of B. Then at least one of A or B has at least 5 elements. Without loss of generality, let $|A|\geq 5$. Then the product of the elements in A, denoted P_A, is greater than or equal to $n(n+1)(n+2)(n+3)(n+4)$. Similarly, $|B|\leq 4$, so the product of the elements in B, denoted P_B, is less than or equal to $(n+5)(n+6)(n+7)(n+8)$. Then we have $n(n+1)(n+2)(n+3)(n+4)\leq P_A=P_B\leq(n+5)(n+6)(n+7)(n+8)$. So the inequality $n(n+1)(n+2)(n+3)(n+4)\leq(n+5)(n+6)(n+7)(n+8)$ must hold. Equivalently, $n+4\leq \left(\frac{n+5}{n}\right)\left(\frac{n+6}{n+1}\right)\left(\frac{n+7}{n+2}\right)\left(\frac{n+8}{n+3}\right)=\left(1+\frac{5}{n}\right)\left(1+\frac{5}{n+1}\right)\left(1+\frac{5}{n+2}\right)\left(1+\frac{5}{n+3}\right)$. The LHS is monotonically increasing as positive n increases and the RHS is monotonically decreasing as positive n increases. The inequality works for 5 and fails for 6, so $n\leq 5$. The set of nine integers cannot contain 0 as 0 can be in at most 1 set, making the product of only that set 0. Further, the set of nine integers cannot only be negative as the size of A and B have opposite parity, and thus their products would have opposite signs. Therefore the only possible sets are $\{1,2,\dots,9\}$, $\{2,3,\dots,10\}$, $\{3,4,\dots,11\}$, $\{4,5,\dots,12\}$, and $\{5,6,\dots,13\}$. Notice that each of these sets have only one multiple of 7. Then the prime factorization of only one of P_A or P_B can have a 7, contradicting the inequality. We conclude that the initial assumption was false, and so the claim follows.

Problem 3. Find the least positive integer n such that any set of n pairwise relatively prime integers greater than 1 and less than 2005 contains at least one prime number.

Solution. Note that the set of numbers $\{2^2, 3^2, 5^2, \ldots, 43^2\}$ are pairwise relatively prime, showing that $n \geq 15$. Assume for contradiction that there exist composite pairwise relatively prime positive integers $m_1, m_2, \ldots, m_{15}$ such that $1 < m_i < 2005$ for all $1 \leq i \leq 15$. Since each of the m_i are composite, there exists a prime factor q_i of m_i such that $q_i \leq \sqrt{m_i} < \sqrt{2005} < 45$. So $q_i \in \{2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43\}$. There are fourteen primes in this set, but we have fifteen integers, so by the Pigeonhole Principle, at leaast one prime is shared by two integers in this set. This contradicts that the integers are relatively prime. Hence our assumption was false and we conclude $n = 15$.

Problem 4. Every point of three-dimensional space is colored red, green, or blue. Prove that one of the colors attains all distances, meaning that any positive real number represents the distance the distance between two points of this color.

Solution.